MAT 3222
Numerical Analysis II

Chapter 1_2: Numerical Solutions to Systems of Linear Equations

Learning Objectives

- Gaussian elimination
- Backward substitution

Cont.

- For example, a system with n equations and m unknowns can be written as:

$$\begin{cases} a_{11}x_1 + a_{12}x_2 + \cdots + a_{1m}x_m = b_1 \\ a_{21}x_1 + a_{22}x_2 + \cdots + a_{2m}x_m = b_2 \\ \vdots \\ a_{n1}x_1 + a_{n2}x_2 + \cdots + a_{nm}x_m = b_n \end{cases}$$

Matrix Form

- The system can be written compactly as:

$$A\mathbf{x} = \mathbf{b}$$

where;

$$A = \begin{bmatrix} a_{11} & a_{12} & \dots & a_{1m} \\ a_{21} & a_{22} & \dots & a_{2m} \\ \vdots & \vdots & \ddots & \vdots \\ a_{n1} & a_{n2} & \dots & a_{nm} \end{bmatrix}, \quad \mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_m \end{bmatrix}, \quad \mathbf{b} = \begin{bmatrix} b_1 \\ b_2 \\ \vdots \\ b_n \end{bmatrix}$$

Triangular Matrix Forms

- **Upper Triangular (U):**

  All elements **below** the **main diagonal** are zero

$$U = \begin{bmatrix} u_{11} & u_{12} & \dots & u_{1m} \\ 0 & u_{22} & \dots & u_{2m} \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & u_{nn} \end{bmatrix}$$

Cont.

- **Lower Triangular (L):**

  All elements **above** the **main diagonal** are zero

$$L = \begin{bmatrix} l_{11} & 0 & \dots & 0 \\ l_{21} & l_{22} & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ l_{n1} & l_{n2} & \dots & l_{nn} \end{bmatrix}$$

- Use in solving systems:
  - Upper triangular → back-substitution
  - Lower triangular → forward-substitution

Augmented Matrix Form

- A matrix that combines the **coefficient matrix** of a linear system and the **constants** into a single matrix.
  - It is used to represent a linear system in a compact form.
  - Helps in applying methods like **Gaussian elimination** or **Gauss-Jordan elimination**.

$$[A|b] = \left[\begin{array}{cccc|c} a_{11} & a_{12} & \dots & a_{1m} & b_1 \\ a_{21} & a_{22} & \dots & a_{2m} & b_2 \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ a_{n1} & a_{n2} & \dots & a_{nm} & b_n \end{array}\right]$$

Gaussian Elimination to Solve Linear Equations

- The Gaussian Elimination Method is a widely used technique for **solving systems of linear equations**, where multiple equations with unknown variables are solved **simultaneously** to find the values of the unknowns.
- This method has practical applications in real life, such as in **traffic flow analysis**, where it helps solve, systems representing traffic movement at intersections, optimizing the flow of traffic and reducing congestion.

Cont.

- In **LU decomposition**, Gaussian elimination is used to **convert** the matrix A into an **upper triangular matrix** (U).

- Note:

LU decomposition expresses a square matrix A as the union of:

$$A = L \cup U$$

where;

- L = Lower triangular matrix (all zeros above diagonal)
- U = Upper triangular matrix (all zeros below diagonal)

Gaussian Elimination Method

- Gaussian elimination is a **row reduction algorithm** for solving linear systems.
- It involves a **series of operations** on the **augmented matrix** (which includes both coefficients and constants) to simplify it into a **row echelon form** or **reduced row echelon form**.
- This method can also help in determining the rank, determinant and inverse of matrices.
- Gaussian elimination is a method for **solving systems of equations in matrix form**.

Cont.

- Let's for example we have system of linear equations:

$$\begin{aligned} a_1x + b_1y + c_1z &= d_1 \\ a_2x + b_2y + c_2z &= d_2 \\ a_3x + b_3y + c_3z &= d_3 \end{aligned}$$

- Matrix form of this system of linear equations:

$$\left[\begin{array}{ccc|c} a_1 & b_1 & c_1 & d_1 \\ a_2 & b_2 & c_2 & d_2 \\ a_3 & b_3 & c_3 & d_3 \end{array}\right]$$

Cont.

- Goal:

  *Turn matrix into **row-echelon form** like,*

Eg:

$$\left[\begin{array}{ccc|c} 0 & a & b & d \\ 0 & 1 & c & e \\ 0 & 0 & 1 & f \end{array}\right]$$

- The entries on the main diagonal may be 1, may be some other non-zero number, or may even be 0.

Elementary Row Operations

- **Interchanging Rows:** Swap two rows.

$$R_i \leftrightarrow R_j$$

- **Multiplying a Row by a Scalar:** Multiply all elements of a row by a non-zero number.

$$R_i \rightarrow 2R_i$$

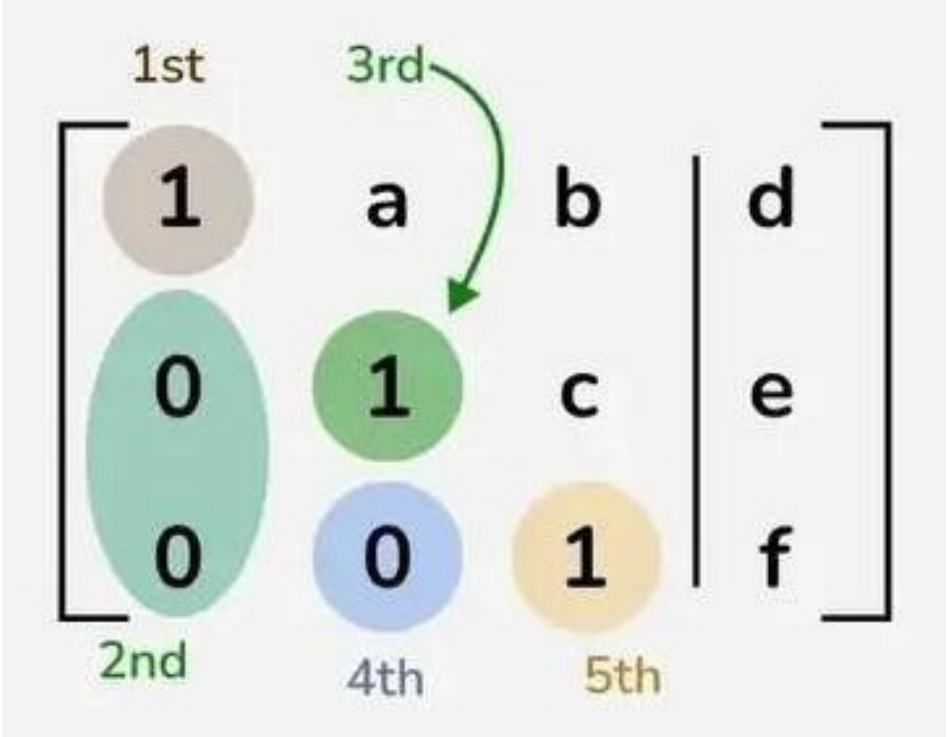
- **Adding a Scalar Multiple of One Row to Another:** Add or subtract a multiple of one row to/from another.

$$R_i \rightarrow R_i + kR_j \text{ or}$$
$$R_i \rightarrow R_i - kR_j$$

- These operations simplify solving systems of linear equations **without changing the solution set**.

Cont.

- Follow these steps:
  1. Get a 1 in the first column, first row.
  2. Use the 1 to get 0's in the remainder of the first column.
  3. Get a 1 in the second column, second row.
  4. Use the 1 to get 0's in the remainder of the second column.
  5. Get a 1 in the third column, third row.



Cont.

- The resulting matrix is in a **row echelon form**.
- A matrix is in **reduced row-echelon form** if **all leading coefficients are 1** and the **columns containing** these leading coefficients have **zeros elsewhere**.
- This final form is unique, regardless of the sequence of row operations used.

General Gauss Elimination Steps

➢Consider an $n \times m$ augmented matrix $[A|b]$.

- Rows = n
- Columns = m
- Pivot steps performed for k=1 to k=min (n, m):

$$[A|b] = \left[\begin{array}{cccc|c} a_{11} & a_{12} & \cdots & a_{1m} & b_1 \\ a_{21} & a_{22} & \cdots & a_{2m} & b_2 \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{nm} & b_n \end{array}\right]$$

Step 1: First Pivot

· Pivot value

$$p_1 = a_{11}$$

· Multiplier for each row below pivot

$$m_{i1} = \frac{a_{i1}}{p_1}$$

· Row operation

$$R_i \to R_i - m_{i1} R_1$$

Cont.

· Updated entries

$$a_{ij}^{(1)} = a_{ij} - m_{i1}a_{1j}$$

$$b_i^{(1)} = b_i - m_{i1}b_1$$

· Matrix after Step 1

$$\left[\begin{array}{ccccc|c} a_{11} & a_{12} & a_{13} & \dots & a_{1m} & b_1 \\ 0 & a_{22}^{(1)} & a_{23}^{(1)} & \dots & a_{2m}^{(1)} & b_2^{(1)} \\ 0 & a_{32}^{(1)} & a_{33}^{(1)} & \dots & a_{3m}^{(1)} & b_3^{(1)} \\ \vdots & \vdots & \vdots & & \vdots & \vdots \\ 0 & a_{n2}^{(1)} & a_{n3}^{(1)} & \dots & a_{nm}^{(1)} & b_n^{(1)} \end{array}\right]$$

Cont.

Step 2: Second Pivot

- Pivot value

$$p_2 = a_{22}^{(1)}$$

- Multiplier

$$m_{i2} = \frac{a_{i2}^{(1)}}{p_2}$$

- Row operation

$$R_i \to R_i - m_{i2} R_2$$

Cont.

· Updated entries

$$a_{ij}^{(2)} = a_{ij}^{(1)} - m_{i2}a_{2j}^{(1)}$$

$$b_i^{(2)} = b_i^{(1)} - m_{i2}b_2^{(1)}$$

· Matrix after Step 2

$$\left[\begin{array}{ccccc|c} a_{11} & a_{12} & a_{13} & \dots & a_{1m} & b_1 \\ 0 & a_{22}^{(1)} & a_{23}^{(1)} & \dots & a_{2m}^{(1)} & b_2^{(1)} \\ 0 & 0 & a_{33}^{(2)} & \dots & a_{3m}^{(2)} & b_3^{(2)} \\ \vdots & \vdots & \vdots & & \vdots & \vdots \\ 0 & 0 & a_{n3}^{(2)} & \dots & a_{nm}^{(2)} & b_n^{(2)} \end{array}\right]$$

Cont.

Step k: Continue pivoting until row k

Pivot coefficient:

$$p_k = a_{kk}^{(k-1)}$$

Multiplier:

$$m_{ik} = \frac{a_{ik}^{(k-1)}}{p_k}$$

Row update:

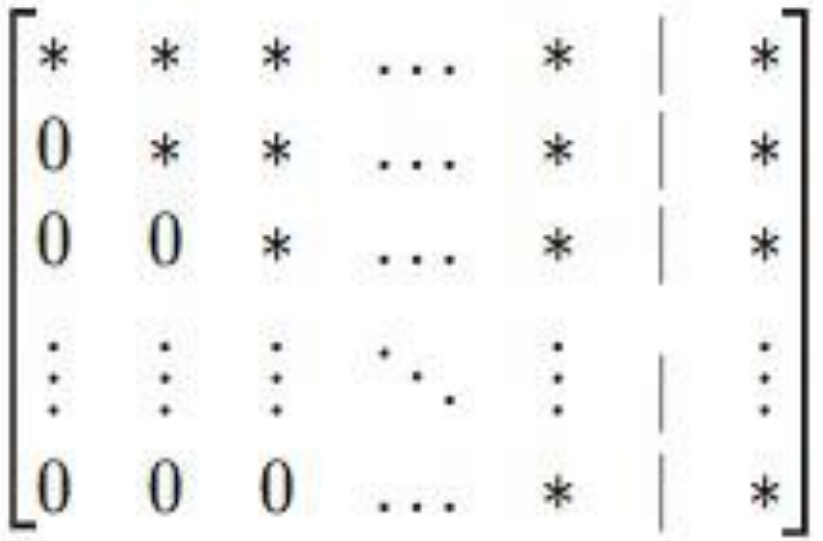
$$a_{ij}^{(k)} = a_{ij}^{(k-1)} - m_{ik} a_{kj}^{(k-1)}$$

$$b_i^{(k)} = b_i^{(k-1)} - m_{ik} b_k^{(k-1)}$$

Cont.

Final Step:

Gaussian elimination ends here:



Then solve using **back-substitution**.

Back Substitution

- Use back-substitution to find the values of the unknowns after obtaining the row echelon form. The steps are:

1. **Start** with the **last row** of the row echelon form.
2. Solve for the **last variable** by dividing the constant term by its coefficient.
3. **Substitute** the **known value** of the last variable **into the previous rows**.
4. **Solve for the next variable** by isolating it and using the known values of the other variables.
5. **Repeat** the process for all rows, **moving upwards** to the first row.
6. Once all variables are solved, the system is complete and the solution is found.

Examples

Example 1:

$$x + y = 3$$
$$3x - 2y = 4$$

Example 2:

$$4x_1 - 2x_2 - 3x_3 + 6x_4 = 12$$
$$-6x_1 + 7x_2 + 6.5x_3 - 6x_4 = -6.5$$
$$x_1 + 7.5x_2 + 6.25x_3 + 5.5x_4 = 16$$
$$-12x_1 + 22x_2 + 15.5x_3 - x_4 = 17$$

Gaussian Elimination with Partial Pivoting

- If a **pivot element is very small**, dividing by it can cause large rounding errors, leading to inaccurate solutions.

- Partial Pivoting Helps:
  - Reduces rounding errors (numerical instability).
  - Avoids division by very small numbers (which can magnify errors).
  - Makes Gaussian elimination more robust for real-world problems.

Cont.

- The **steps** involved in Partial Pivoting Gaussian Elimination are:

1. Start with the given augmented matrix.
2. Find **the row with the largest absolute value in the first column** and **swap that row with the first row**.
3. Divide the first row by the pivot element to make it equal to 1.
4. Use the first row to eliminate the first column in all other rows by subtracting the appropriate multiple of the first row from each row.
5. Repeat steps 2-4 for the remaining columns, using the row with the largest absolute value as the pivot row for each column.
6. Back-substitute to obtain the values of the unknowns.

Example

Example 3:

$$0.003x + 59.14y = 59.17$$
$$5.291x - 6.13y = 46.78$$

Gauss-Jordan Elimination

- A method to solve linear equations by transforming the coefficient matrix into reduced **row-echelon form** (**diagonal form**) using row operations.
- Key Feature:
  - Eliminates all entries above and below the pivots.
  - Solution can be read directly from the matrix (no back-substitution needed).

Cont.

Gauss-Jordan Elimination Steps:

1. **Pivot Selection:** Make the pivot 1 (usually a_{ii}) using row operations.
2. **Eliminate below and above pivot:** Make all other entries in the pivot column 0.
3. **Repeat** for all pivot columns.

Resulting Form (Reduced Row-Echelon Form):

$$\left[\begin{array}{ccccc|c} 1 & 0 & 0 & \cdots & 0 & x_1 \\ 0 & 1 & 0 & \cdots & 0 & x_2 \\ 0 & 0 & 1 & \cdots & 0 & x_3 \\ \vdots & \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & 0 & \cdots & 1 & x_n \end{array}\right]$$

Example

Example 4:

$$x + y + z = 6$$
$$2x + 3y + z = 10$$
$$x + 4y + 5z = 20$$

Applications of Gaussian Elimination Method

- **Determinants:** Simplifies calculation by reducing to row echelon form; determinant = product of diagonal elements (adjusted for row swaps and scalars).
- **Matrix Inverse:** Augment with identity matrix and apply row operations to obtain the inverse (if it exists).
- **Ranks & Bases:** Row echelon form reveals rank and identifies linearly independent columns forming a basis.